EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The Discrete-Time Fourier Transform (DTFT) represents discrete sequences in the continuous frequency domain $\omega \in [-\pi, \pi]$. It provides the frequency analysis tool for aperiodic discrete-time signals and forms the theoretical bridge to filter frequency response and the DFT.

Pedagogical Strategy:

1. Derive the DTFT from continuous-time Fourier analysis of sampled impulse trains.
2. Emphasize the inherent 2π-periodicity of the DTFT: $X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$.
3. Rigorously define existence/convergence: The DTFT converges uniformly if $x[n]$ is absolutely summable ($\sum |x[n]| < \infty$).
4. Systematically derive the fundamental theorems: Linearity, Time-Shifting, Frequency-Shifting (Modulation), Differentiation in Frequency, Time Convolution, and Parseval's Energy Theorem.
5. Highlight Hermitian symmetry for real sequences: Magnitude is an even function ($|X(e^{-j\omega})| = |X(e^{j\omega})|$), phase is an odd function ($\angle X(e^{-j\omega}) = -\angle X(e^{j\omega})$).

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Compute** the forward DTFT $X(e^{j\omega})$ and Inverse DTFT (IDTFT) $x[n]$ for standard and composite signals.
2. **Apply** DTFT transform properties to evaluate complex frequency responses without direct integration.
3. **Analyze** symmetry properties of $X(e^{j\omega})$ for real, imaginary, even, and odd sequences.
4. **Use** Parseval's Theorem to compute total energy in both time and frequency domains.
5. **Differentiate** between the DTFT (continuous frequency) and the DFT (sampled discrete frequency).

2. MATHEMATICAL FOUNDATIONS

2.1 The DTFT Analysis and Synthesis Equations

- **Forward DTFT (Analysis Equation):**

$$X(e^{j\omega}) = \mathcal{F}x[n] = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- **Inverse DTFT (Synthesis Equation):**

$$x[n] = \mathcal{F}^{-1}X(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

2.2 Convergence Condition

The infinite series converges uniformly to a continuous function of ω if $x[n]$ is absolutely summable (sufficient condition):

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

If $x[n]$ is square-summable ($\sum |x[n]|^2 < \infty$), the DTFT converges in the mean-square sense.

2.3 Comprehensive Table of DTFT Properties

Let $x[n] \leftrightarrow X(e^{j\omega})$ and $y[n] \leftrightarrow Y(e^{j\omega})$:

Property	Time Domain	Frequency Domain
Periodicity	$x[n]$	$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$
Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
Time Shifting	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
Frequency Shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega-\omega_0)})$
Time Reversal	$x[-n]$	$X(e^{-j\omega})$
Differentiation in Freq	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$
Convolution in Time	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
Multiplication in Time	$x[n] \cdot y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$
Parseval's Relation	$\sum_{n=-\infty}^\infty	x[n]

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3. WORKED NUMERICAL EXAMPLES

Example 3.1: DTFT of Exponential Decaying Sequence

Problem: Find the DTFT of $x[n] = a^n u[n]$ where $|a| < 1$. Find magnitude and phase.

Solution:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} a^n u[n] e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

Since $|ae^{-j\omega}| = |a| < 1$, using geometric series $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$:

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}} = \frac{1}{1 - a\cos\omega + ja\sin\omega}$$

- **Magnitude Response:**

$$|X(e^{j\omega})| = \frac{1}{\sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2}} = \frac{1}{\sqrt{1 - 2a\cos\omega + a^2}}$$

- **Phase Response:**

$$\angle X(e^{j\omega}) = -\arctan\left(\frac{a\sin\omega}{1 - a\cos\omega}\right)$$

Example 3.2: Inverse DTFT of an Ideal Lowpass Filter

Problem: Find the impulse response $h_d[n]$ of an ideal Lowpass filter with cutoff ω_c:

$$H_d(e^{j\omega}) = \{1, \quad |\omega| \leq \omega_c\ 0, \quad \omega_c < |\omega| \leq \pi$$

Solution: Using the IDTFT synthesis integral:

$$h_d[n] = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega = \frac{1}{2\pi jn} e^{j\omega n}\Big|_{-\omega_c}^{\omega_c} = \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{2\pi jn} = \frac{\sin(\omega_c n)}{\pi n}$$

For $n = 0$, applying L'Hôpital's rule: $h_d[0] = \frac{\omega_c}{\pi}$. Thus:

$$h_d[n] = \frac{\sin(\omega_c n)}{\pi n} = \frac{\omega_c}{\pi}\operatorname{sinc}\left(\frac{\omega_c n}{\pi}\right)$$

Note: $h_d[n]$ is infinite in duration and non-causal (since $h_d[n] \neq 0$ for $n < 0$), making ideal brick-wall filters physically unrealizable.

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State and prove the Time-Shifting and Convolution properties of the DTFT. *(6 Marks)* **(b)** Using DTFT properties, find the frequency response $X(e^{j\omega})$ of $x[n] = (n+1)a^n u[n]$ for $|a| < 1$. *(5 Marks)* **(c)** Evaluate the total energy E of the sequence $x[n] = \frac{\sin(\pi n/3)}{\pi n}$ using Parseval's Theorem. *(4 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Time Shift proof: $\mathcal{F}x[n-n_0] = \sum x[n-n_0]e^{-j\omega n}$. Let $m = n - n_0 \implies e^{-j\omega(m+n_0)} = e^{-j\omega n_0}X(e^{j\omega})$ *(3 Marks)*
  - Convolution proof: $\mathcal{F}x * y = \sum_n [\sum_k x[k]y[n-k]]e^{-j\omega n} = \sum_k x[k]e^{-j\omega k}\sum_m y[m]e^{-j\omega m} = X(e^{j\omega})Y(e^{j\omega})$ *(3 Marks)*
- **Part (b):**
  - Let $v[n] = a^n u[n] \leftrightarrow V(e^{j\omega}) = \frac{1}{1-ae^{-j\omega}}$.
  - $x[n] = nv[n] + v[n]$.
  - Applying differentiation in frequency: $\mathcal{F}nv[n] = j\frac{d}{d\omega}\left(\frac{1}{1-ae^{-j\omega}}\right) = j\frac{-(-jae^{-j\omega})}{(1-ae^{-j\omega})^2} = \frac{ae^{-j\omega}}{(1-ae^{-j\omega})^2}$.
  - $X(e^{j\omega}) = \frac{ae^{-j\omega}}{(1-ae^{-j\omega})^2} + \frac{1}{1-ae^{-j\omega}} = \frac{1}{(1-ae^{-j\omega})^2}$ *(5 Marks)*
- **Part (c):**
  - By Parseval's theorem: $E = \sum |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$.
  - Here $X(e^{j\omega})$ is an ideal pulse of height 1 over $[-\pi/3, \pi/3]$.
  - $E = \frac{1}{2\pi}\int_{-\pi/3}^{\pi/3} 1^2 d\omega = \frac{1}{2\pi}\left(\frac{2\pi}{3}\right) = \frac{1}{3}$ Joules. *(4 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```python
import numpy as np

# Energy verification via Parseval's
n = np.arange(-1000, 1001)
# Handle n=0 limit
x = np.zeros_like(n, dtype=float)
x[n != 0] = np.sin(np.pi * n[n != 0] / 3) / (np.pi * n[n != 0])
x[n == 0] = 1.0 / 3.0

E_time = np.sum(x**2)
E_freq = 1.0 / 3.0
print(f"Time-domain Energy = {E_time:.6f}, Theoretical = {E_freq:.6f}")
```